

5.3 Retrieval, Bias, and Robustness

recall@k in both directions, an occupation probe, and a typographic miss.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

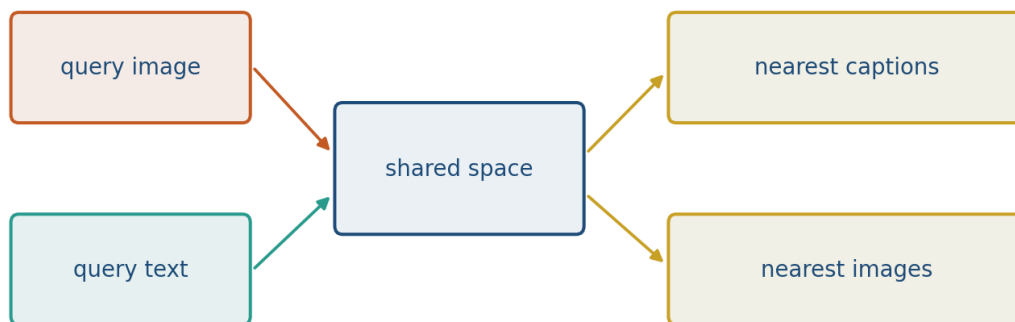
Once image and text share a space, **retrieval** is nearest neighbors. The same geometry that makes search work also encodes **stereotypes** from the training web. A single prompt can be peaked; this note measures what that geometry retrieves. Lab 5's occupation probe is the miniature; the lecture is the metric and the probe.

1. What retrieval metrics and bias probes are

Recall@k is a retrieval metric: the fraction of queries whose true match sits in the top k . **Bias probes** and **typographic attacks** are how you stress a CLIP-like space. An occupation probe asks whether "nurse" retrieves stereotyped images. A typographic attack prints a word in the photo and hijacks the text tower.

Text → **image**: a query string, rank photos. **Image** → **text**: a photo, rank captions or documents. Dual retrieval is why coordinated towers beat a joint vector that you can only query from one side.

One space, two retrieval directions



If the gold item is in the top k of the ranked list, that query scores 1; otherwise 0. Average over queries. That is recall@k. It is cheap, and it does not measure calibration or fairness by itself.

2. Why we use it

A pretty cosine is not a fair or robust system. You need a number that can go the wrong way, and a probe set that can embarrass a demo.

Recall@1 on a 2-item toy is not a paper. On a 100k gallery, recall@1 can look terrible while recall@10 is usable. Always state k and gallery size. For a project, freeze the gallery embeddings; do not re-encode millions of images on every demo.

If your project ranks people, jobs, or health content, put a probe in the report: a small, documented set of prompts and the retrievals they return. Debiasing is incomplete; measuring is required. A high ImageNet zero-shot number does not certify your domain.

3. Architecture

Same two CLIP towers. Evaluation is a ranked list plus extra probe sets. You do not add a new layer. You add a gallery, a query encoder, and a rule for a hit.

Keep both towers. Throw one away and you can no longer query from that side. Rank **both** directions (image $\rightarrow$ text and text $\rightarrow$ image). Report recall@1 and recall@5, then one occupation table and one typographic miss.

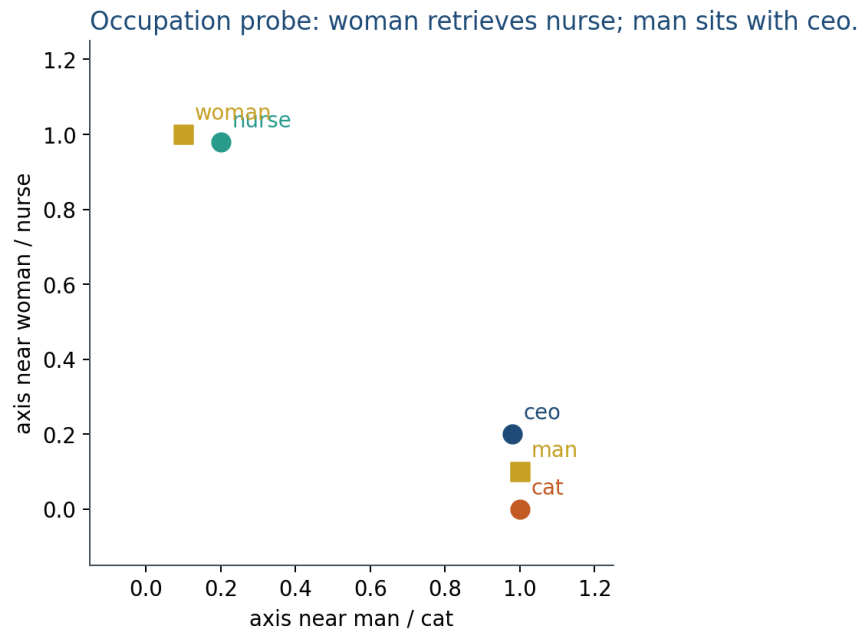
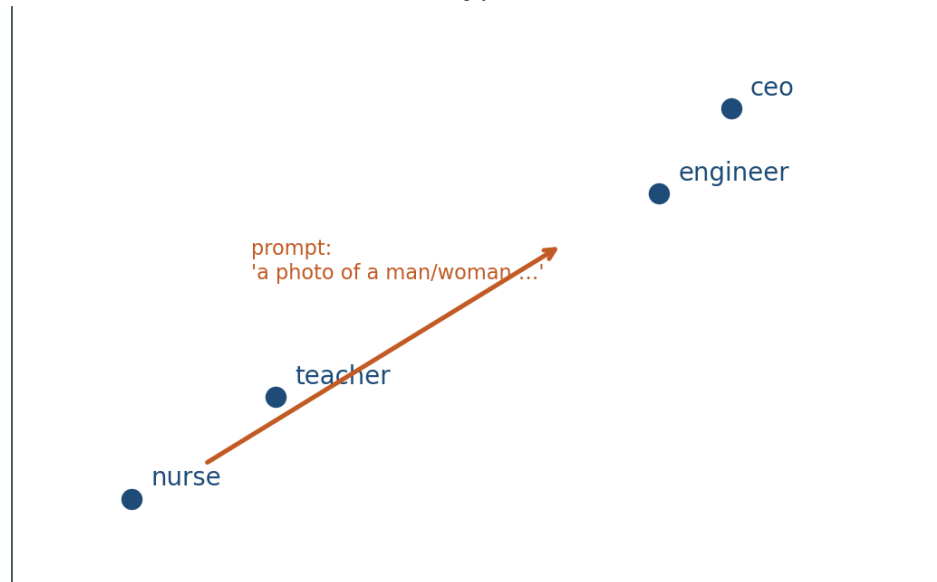
A probe is a **fixed list** of prompts and a rule for a bad hit (wrong gender default, violent stereotype, medical advice). Put the list in an appendix. Do not “debias” by deleting one axis and claiming the space is fair; say what you measured.

Lab 5’s occupation probe (nurse / ceo vs man / woman in toy_clip.csv) is the miniature of this architecture. Real CLIP is the same cosine, trained on the web.

4. How it works, step by step

1. **Embed the gallery** once. Store vectors plus ids plus the raw items you will show.
2. **Embed the query** with the matching tower (text for a typed query, image for reverse search).
3. **Rank** by cosine. For each query, mark whether gold sits at rank $\leq k$. Average: that is recall@k.
4. **Read the misses.** Wrong object, wrong count, text that ignores the image. A few failure cases belong next to the number.
5. **Run a bias probe.** Occupation and gender, geography and skin tone, “a terrorist” versus a news photo: CLIP-style models pick up the co-occurrences of the web. Zero-shot prompts (a photo of a man who is a nurse) move the query along the same directions you saw in Week 1’s toy embeddings.
6. **Run a robustness check.** Typographic attacks (text painted in the image), distribution shift (medical, sketches), and language variety.

VLMs inherit dataset stereotypes in retrieval and zero-shot



Goh et al. / the CLIP "multimodal neurons" line: a photo of an apple with the word `iPod` printed on it is often retrieved as a gadget. CLIP can **read the overlay** instead of the object. That is the same nearest-neighbor geometry, not a separate model.

Typographic attack: CLIP often reads the printed word first.



If the overlay wins, write that failure next to the occupation table. Two different harms, one cosine.

5. Mathematical formulas

For Q queries, with $\text{rank}(q)$ the position of the gold item (1 is best),

$$\text{recall}@k = \frac{1}{Q} \sum_q 1[\text{rank}(q) \leq k].$$

Cosine ranking is the same score as CLIP:

$$\text{rank}(q) = \left| \{z: \cos(e(q), e(z)) > \cos(e(q), e(z_{\text{gold}}))\} \right| + 1$$

(ties: pick a rule and stick to it). Always name k and the gallery size when you report the number.

6. Positive points and negative points

Positive.

- Recall@k is cheap to compute and catches misses before a demo ships.
- Dual retrieval is the reason you kept both towers.
- A documented probe list is a method; Lab 5's occupation rows are the template.

Negative.

- Recall@k does not measure calibration or fairness by itself.
- Probes are datasets, not slogans: you need a fixed list and a rule for a bad hit.
- A typographic overlay can hijack the text tower; that is the geometry working as designed.
- Reporting a single recall@1 without k or gallery size hides usable rank-3 hits.

When not to. Do not skip a bias probe because “it’s just a toy.” Do not treat a typographic miss as “the model is broken”; text in the image is also a vector.

7. Teaching this note

30–40 minutes. Dual retrieval, recall@k with a tiny ranking, then the occupation probe **with numbers**, then one typographic cartoon. Play the **second half** of the CLIP video (robustness / broader impact, about **37:40–47:00**). Mention Lab 5’s occupation rows by name so the lab is not a surprise.

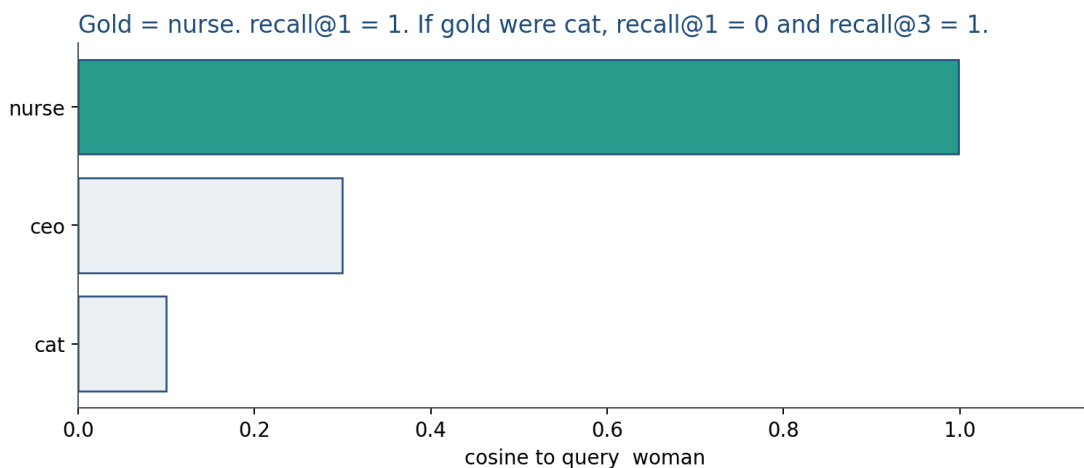
Minute plan: 8 min two retrieval directions; 10 min recall@k arithmetic; 10 min occupation probe; 10 min video second half. Do not skip the probe to “save time.”

8. Worked example

Gallery of three ℓ_2 -normalized image vectors:

$$\text{cat} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \text{ nurse} = \begin{bmatrix} 0.2 \\ 0.98 \end{bmatrix}, \text{ ceo} = \begin{bmatrix} 0.98 \\ 0.2 \end{bmatrix}.$$

Query text woman $\approx \begin{bmatrix} 0.1 \\ 1 \end{bmatrix}$. Cosines $\approx 0.10, 0.999, 0.30$. Rank: nurse, ceo, cat.



Query text man $\approx \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}$. Cosines $\approx 0.995, 0.30, 0.995$. Rank: ceo tied with cat; nurse last.

That is an **occupation probe**: who is nearest to `man` vs `woman`. Lab 5 asks you to run it on `toy_clip.csv`. A production system that ranks résumés needs the same table in the report, on real photos, with documented prompts.

Recall@1 for query `cat` is 1 if cat is the top image. If you only report recall@1 on 100k images, a model that puts the right photo at rank 3 looks like a zero.

Typographic cartoon: query prompts `apple` vs `iPod` against one image. If the overlay wins, write that failure next to the occupation table. Two different harms, one cosine.

9. Where students get stuck

- Throwing away one tower and then being unable to query from that side.
 - Reporting a single recall@1 without k or gallery size.
 - Skipping a bias probe because "it's just a toy."
 - Treating a typographic miss as "the model is broken" instead of "text in the image is also a vector."
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10. Video

Watch the **second half** of [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Start around **robustness to data shift (37:40)** and **broader impact (44:20)**. Pause on prompt sensitivity and on who is harmed when retrieval ranks people. Pair that with Lab 5's occupation probes (nurse / ceo vs man / woman), not with a vibe check.

11. Practice

1. You can retrieve with a caption but not with a photo. What did you probably throw away?
2. Why is recall@1 a poor single number for a 100k image gallery?
3. Write one probe prompt you would be embarrassed to ship, and what you would count as a bad retrieval.
4. Cosines of a query to four images: 0.11, 0.40, 0.39, 0.38. If gold is the 0.40 image, what is recall@1 and recall@3? If gold is the 0.38 image, write both numbers.
5. Vectors $q = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $a = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$, $b = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$. Rank a, b by cosine to q . Then ℓ_2 -normalize a and b and rank again. Did the order change? Why or why not?